15C+16
CHEMICAL EQUILIBRIUM

UNIT 02: HOW REACTIONS HAPPEN

CHEMISTRY 1310

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LEARNING OBJECTIVES

CHAPTER 15

1. Define and explain the condition of equilibrium.
2. Determine equilibrium constant expressions and their values.
3. Use equilibrium constant expressions as formulas.
4. Manipulate equilibrium constant expressions based on changes to the chemical equation.
5. Define and explain the reaction quotient.
6. Predict the direction of a reaction using the relationship between the reaction quotient and the equilibrium constant.
7. Use ICE tables to solve equilibrium problems.
8. Identify the reaction conditions that shift the equilibrium position of a reaction.

CHAPTER 16

1. Write ionization reactions for strong and weak acids and bases.
2. Calculate the hydrogen ion concentration in solutions of strong acids and the hydroxide ion concentration in solutions of strong bases.
3. Identify conjugate acid-base pairs using the Brønsted–Lowry definitions of acids and bases.
4. Interpret the relationship between K_a or K_b on the concentrations of species in acidic or basic solutions.
5. Calculate the pH of a solution.

Le Châtelier's Principle

15.6

When a system is at equilibrium and the conditions that affect the equilibrium are changed (perturbed), the rates of the forward and reverse reactions may be affected *differently*.

$$aA + bB \rightleftharpoons cC + dD$$

$$K = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

If $\mathrm{Rate_{for}} \neq \mathrm{Rate_{rev}}$, more reactants or more products will be produced until the *correct* ratio of products to reactants, governed by K, is re-established.

Systems will always tend toward equilibrium (K)!

Le Châtelier's Principle

- If a *stress* is applied to a system at equilibrium, the reaction will tend to *shift* in a direction to relieve the stress and re-establish equilibrium.
- Le Châtelier's principle is based on the comparison of Q to K.

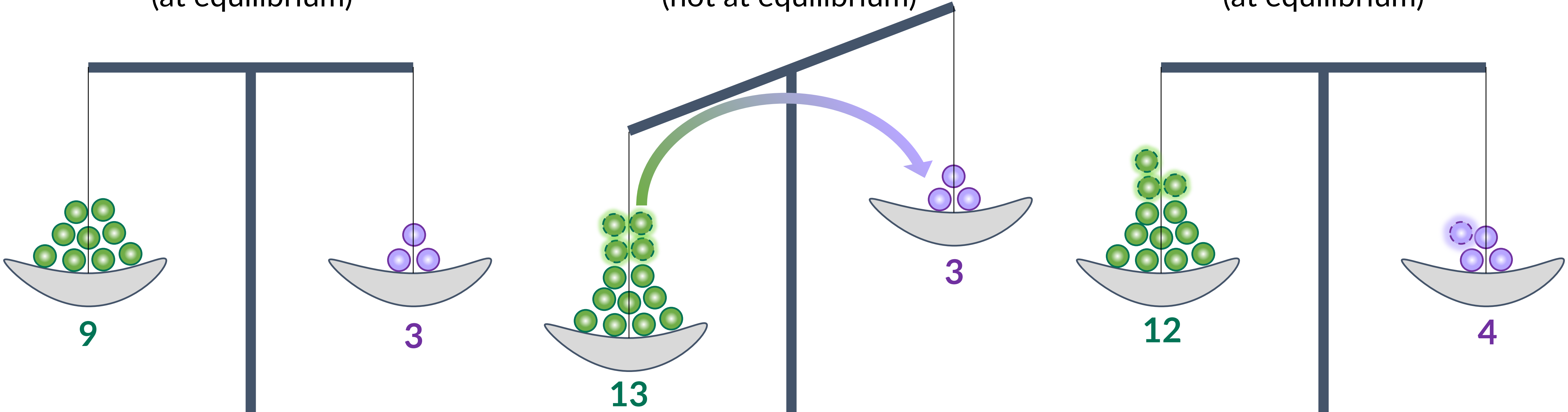
Le Châtelier's Principle

15.6

The stress must affect species in expression of K, which defines the equilibrium state, in order for the system to respond. Otherwise, equilibrium unperturbed. For example,

reactants $\rightleftharpoons$ products $[A] \rightleftharpoons [B]$ $$K = \frac{[B]_{eq}}{[A]_{eq}} = 0.33$$

$Q = K$ (at equilibrium) $Q < K$ (not at equilibrium) $Q = K$ (at equilibrium)



Le Châtelier's Principle

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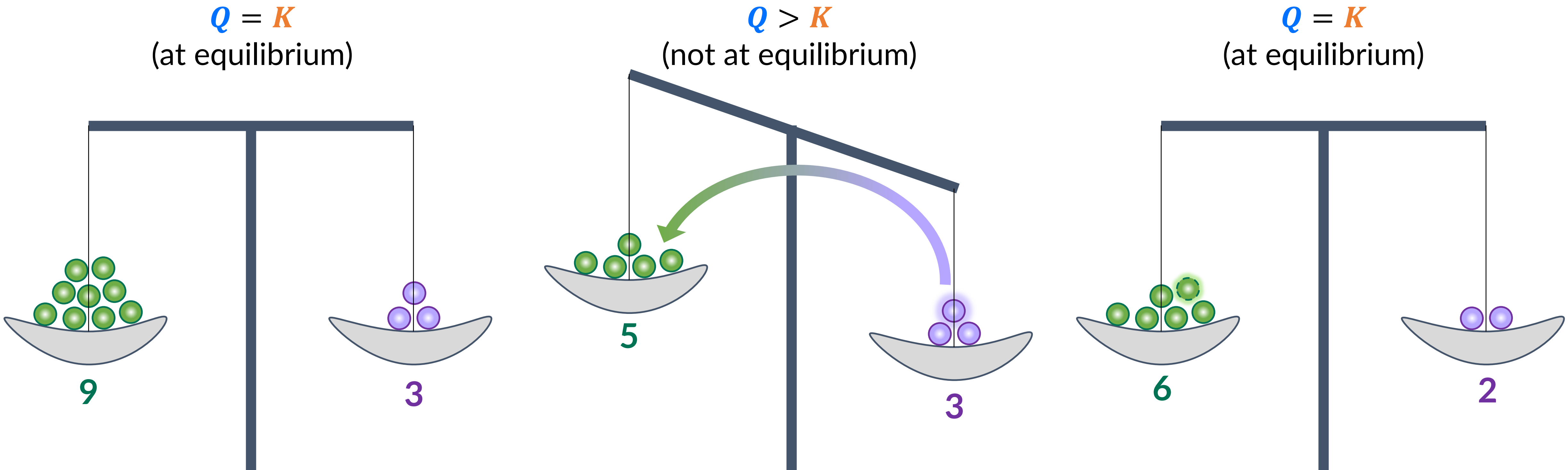


Le Châtelier's Principle

15.6

Le Châtelier's Principle: Effect of reactant/product quantities

- If a *stress* is applied to a system at equilibrium, the reaction will tend to *shift* in a direction to relieve the stress and re-establish equilibrium.
- Le Châtelier's principle is based on the comparison of Q to K. reactants $\rightleftharpoons$ products

	Change in amount	Q vs. K	Reaction shift	Result
Reactant	Increase	$Q < K$	$\longrightarrow$ right or forward	Consumes excess reactant; Forms *more products*
	Decrease	$Q > K$	$\longleftarrow$ left or reverse	Consumes products; Forms *more reactants*
Product	Increase	$Q > K$	$\longleftarrow$ left or reverse	Consumes excess products; Forms *more reactants*
	Decrease	$Q < K$	$\longrightarrow$ right or forward	Consumes reactant; Forms *more products*

Le Châtelier's Principle

15.6

Changes in pressure

- If pressure is *increased*, the reaction will shift in a way to *reduce* overall pressure.
- If pressure is *decreased*, the reaction will shift in a way to *increase* overall pressure.

What does this mean in terms of left and right shifts?

- Only consider gaseous species when dealing with pressure changes.
- Look at the balanced chemical equation to determine which side (reactants or products) has more and which has less gases.
- Recall that number of moles, pressures, and volumes of gases are related. $P \propto n \propto \frac{1}{V}$
  a) Shifting to the *side with more gases* increases the pressure.
  b) Shifting to the *side with less gases* decreases the pressure.

Le Châtelier's Principle

15.6

Changes in pressure

- If pressure is *increased*, the reaction will shift in a way to *reduce* overall pressure.
- If pressure is *decreased*, the reaction will shift in a way to *increase* overall pressure.

$P \propto n \propto \frac{1}{V}$

2 mol gas reactants   1 mol gas products

$$2\,NO_2(g) \rightleftharpoons 1\,N_2O_4(g)$$

Compression → Volume reduced → Pressure increased   e.g.

- Reaction shifts to side with ***less moles of gas***.

Expansion → Volume increased → Pressure decreased

- Reaction shifts to side with ***more moles of gas***.

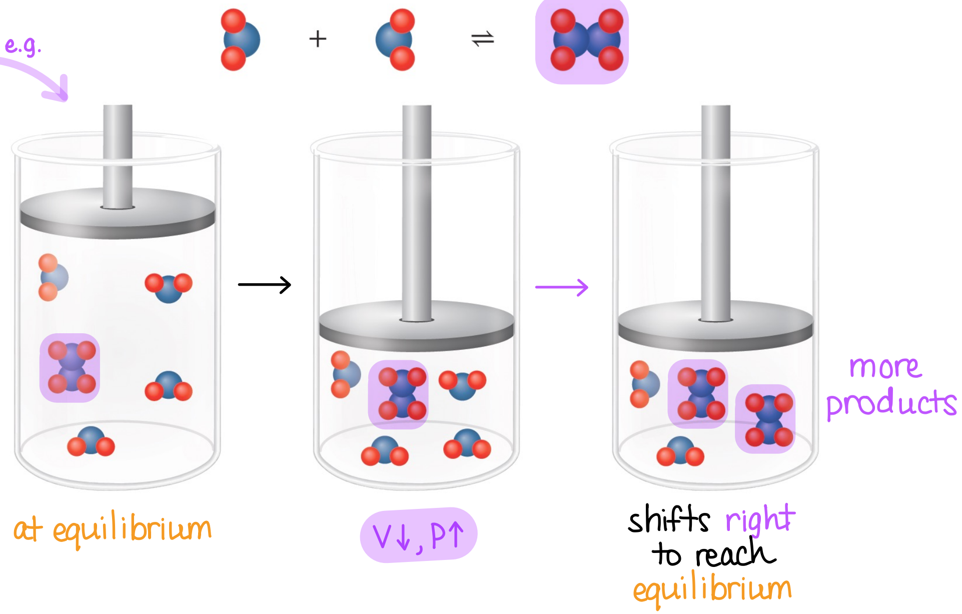


FIGURE 15.7

Le Châtelier's Principle

15.6

What is actually happening?

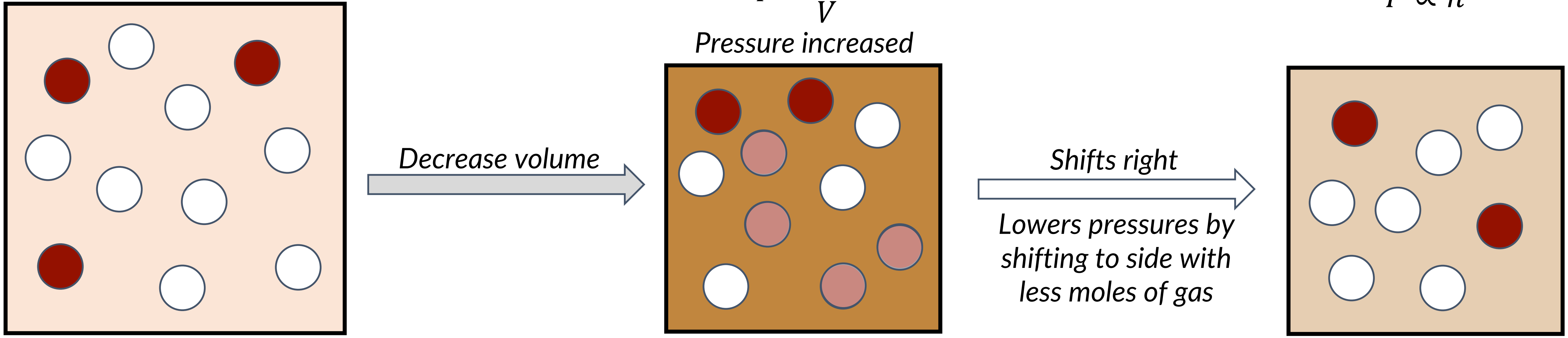
$$2\ NO_2\ (g) \rightleftharpoons 1\ N_2O_4\ (g)$$

$$K_p = \frac{P_{N_2O_4}}{\left(P_{NO_2}\right)^2}$$

$P \propto \frac{1}{V}$

Pressure increased

$P \propto n$



$Q = K$ (at equilibrium)

$Q < K$ (not at equilibrium)

$Q = K$ (at equilibrium)

Le Châtelier's Principle

15.6

The effect of pressure and/or volume

- The net change in moles of gas (Δn_{gas}) should be used to determine shifts.

$$\Delta n_{\text{gas}} = n_{\text{gas products}} - n_{\text{gas reactants}}$$

these are stoichiometric coefficients from balanced chemical equation

Δn_{gas}	Change in volume	Change in pressure	Reaction shift	Result
$\Delta n_{\text{gas}} < 0$	Expansion	decrease	$\longleftarrow$ left or reverse	Toward side with *more gas*
	Compression	increase	$\longrightarrow$ right or forward	Toward side with *less gas*
$\Delta n_{\text{gas}} = 0$	Either	Either	No shift	No change
$\Delta n_{\text{gas}} > 0$	Expansion	decrease	$\longrightarrow$ right or forward	Toward side with *more gas*
	Compression	increase	$\longleftarrow$ left or reverse	Toward side with *less gas*

IN-CLASS QUESTION 1

Learning Objective(s): 15.8

reactants $\rightleftharpoons$ products

Select all changes that will cause a shift toward the ***right*** for the following reaction?

$$2\ NaHCO_3(s) \rightleftharpoons Na_2CO_3(s) + CO_2(g) + H_2O(g) \qquad \Delta n_{gas} = 2 - 0 = 2$$

- ☒ Adding $H_2O(g)$ shifts left ; $Q > K$
- ☒ Removing some but not all $NaHCO_3(s)$ no shift/effect because solids/liquids never in $K = [CO_2][H_2O]$
- ☒ Adding $Na_2CO_3(s)$ no shift/effect because solids/liquids never in $K = [CO_2][H_2O]$
- ☑ Removing $CO_2(g)$ shifts right ; $Q < K$
- ☑ Decreasing the overall pressure shifts right because it's side with more gas $\Delta n_{gas} = 2 - 0 = +2$
- ☒ Decreasing the total volume decreasing volume increases pressure, so rxn shifts left because it's side with less gas

What does a catalyst do?

Chemistry in Context

The presence of a catalyst does *not* affect $\boldsymbol{Q}$ or $\boldsymbol{K}$.

At equilibrium, the following are true:

$$\mathrm{Rate}_{\mathrm{for}} = \mathrm{Rate}_{\mathrm{rev}}$$

$$\boldsymbol{K} = \frac{k_{\mathrm{for}}}{k_{\mathrm{rev}}}$$

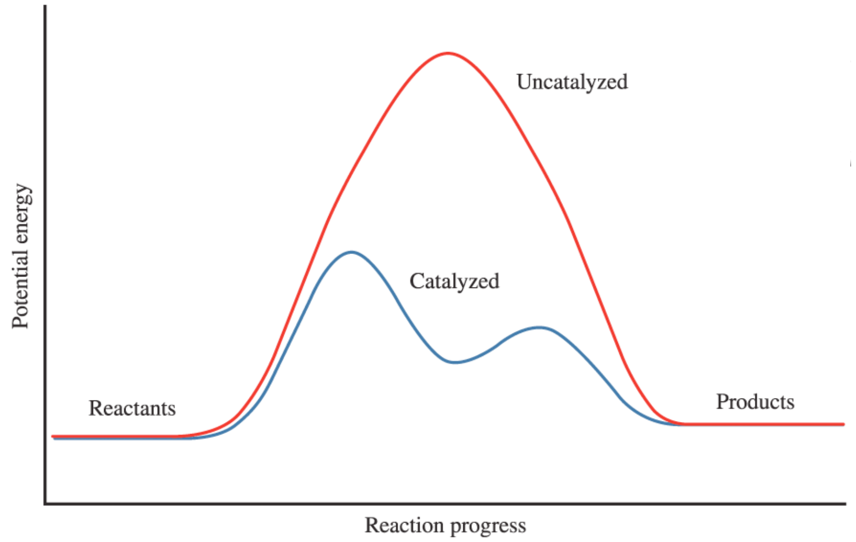


FIGURE 15.11

A catalyst lowers the activation energy for both the forward and reverse reactions *by the same amount*.

The effect of this is that ***equilibrium is established more quickly*** in the presence of a catalyst, but the equilibrium position itself (defined by $\boldsymbol{K}$) is *unchanged*.

Ionization Reactions of Acids and Bases

16.1

Arrhenius acid-base theory: The first modern definition of acids and bases reliant on H^+ for acids and OH^- for bases. Only aqueous solutions where water is the solvent.

- Arrhenius acids produce H^+ when dissolved in water.

$$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$$

- Arrhenius bases produce hydroxide ions (OH^-) when dissolved in water.

$$NaOH(aq) \rightarrow Na^+(aq) + OH^-(aq)$$

What chemists call a "proton"?

Chemistry in Context

A neutral hydrogen atom (H) has 1 proton, 0 neutrons, and 1 electron.

$$\mathrm{H} \xrightarrow{\text{lose } 1\,e^-} \mathrm{H^+}$$

A cationic hydrogen ion (H^+) has 1 proton, 0 neutrons, and 0 electrons.

- This is why H^+ is referred to, simply, as a "***proton***."

Protons (H^+ ions) are *highly reactive* and **do not** exist as H^+ in *aqueous* solutions.

- H^+ will attach to (**protonate**) a water molecule to form a **hydronium ion** (H_3O^+).

$$H^+(aq) + H_2O(\ell) \rightarrow H_3O^+(aq)$$

For our purposes, these are synonymous: proton = H^+ = H_3O^+ = hydronium ion

Brønsted-Lowry Theory

16.2

Brønsted-Lowry acid-base theory: A second definition of acids and bases that is based *only* on proton (hydrogen ions or H^+ ions) behavior.

- **Brønsted-Lowry acids** (HA) are proton (H^+) *donors*.
- **Brønsted-Lowry bases** (B) are proton (H^+) *acceptors*.

$$\underset{\text{acid}}{HA} + \underset{\text{base}}{B} \longrightarrow A^- + HB^+$$

transfer H^+

This is the definition most commonly used today.

Brønsted-Lowry Theory

16.2

A fundamental principle of Brønsted-Lowry theory is that, if a proton donor (an acid, HA) is present then a proton acceptor (a base, B) must also be present.

- Therefore, the reaction between an acid and base is a **proton transfer reaction**.

Conjugate acid-base pairs differ *only* by the presence or absence of a single proton (H^+).

acid base

$$HA + B \longrightarrow A^- + HB^+$$

transfer H^+

acid base conjugate base conjugate acid

$$HCl + NH_3 \longrightarrow Cl^- + NH_4^+$$

transfer H^+

EXAMPLE PROBLEM

Learning Objective(s): 16.3

For each species, identify the conjugate acid or the conjugate base.

The conjugate acid of HCO_3^-

$$\underset{\text{base}}{HCO_3^-} \xrightarrow{\text{add } H^+} \underset{\text{conjugate acid}}{H_2CO_3}$$

The conjugate base of HNO_3

$$\underset{\text{acid}}{HNO_3} \xrightarrow{\text{remove } H^+} \underset{\text{conjugate base}}{NO_3^-}$$

The conjugate acid of HPO_4^{2-}

$$\underset{\text{base}}{HPO_4^{2-}} \xrightarrow{\text{add } H^+} \underset{\text{conjugate acid}}{H_2PO_4^-}$$

Ionization Reactions of Acids and Bases

16.1

Chemical formulas for acids typically begin with H to indicate that they ionize (or dissociate) when dissolved in water.

A strong acid is a strong electrolyte and dissociates/ionizes 100 % in water.

- As the ionization is *complete*, this is *not* an equilibrium.
- There are seven strong acids. All other acids are weak acids.

Strong Acid Name	Formula	Ions in Aqueous Solution	
Hydrochloric acid	HCl	H^+	Cl^-
Hydrobromic acid	HBr	H^+	Br^-
Hydroiodic acid	HI	H^+	I^-
Nitric acid	HNO_3	H^+	NO_3^-
Perchloric acid	$HClO_4$	H^+	ClO_4^-
Chloric acid[1]	$HClO_3$	H^+	ClO_3^-
Sulfuric acid[2]	H_2SO_4	H^+	HSO_4^-

Notes:

1. Some sources consider chloric acid to be a weak acid.
2. Only one of the H atoms completely dissociates. In other words, H_2SO_4 is a strong acid but HSO_4^- is a weak acid.

Ionization Reactions of Acids and Bases

16.1

Chemical formulas for bases typically end with OH (contain hydroxide ions), or end with an H to differentiate them from acids.

A **strong base** is a strong electrolyte and dissociates/ionizes 100 % in water.

- As the ionization is *complete*, this is *not* an equilibrium.
- There are eight strong bases.
- All other bases are weak bases.

Strong Base Name	Formula	Ions in Aqueous Solution	
Lithium hydroxide	$LiOH$	Li^+	OH^-
Sodium hydroxide	$NaOH$	Na^+	OH^-
Potassium hydroxide	KOH	K^+	OH^-
Rubidium hydroxide	$RbOH$	Rb^+	OH^-
Cesium hydroxide	$CsOH$	Cs^+	OH^-
Calcium hydroxide*	$Ca(OH)_2$	Ca^{2+}	OH^-, OH^-
Strontium hydroxide*	$Sr(OH)_2$	Sa^{2+}	OH^-, OH^-
Barium hydroxide*	$Ba(OH)_2$	Ba^{2+}	OH^-, OH^-

* Note: When $Ca(OH)_2$, $Sr(OH)_2$, and $Ba(OH)_2$ dissociate in water, they yield <u>two</u> OH^- anions per unit of strong base.

Ionization Reactions of Acids and Bases

16.1

A <u>weak acid</u> is a weak electrolyte and dissociates/*ionizes incompletely* in water.

- As the ionization is *incomplete* (often <10 %) this is an ***equilibrium*** process.

only H_3O^+, Cl^-
no HCl!

strong acid $\quad HCl + H_2O \xrightarrow{100\%} H_3O^+ + Cl^- \quad K_a \gg 1$

mostly HF!
few H_3O^+, F^-

weak acid $\quad HF + H_2O \xrightleftharpoons{<10\%} H_3O^+ + F^- \quad K_a < 1$

- Given the same number of moles of dissolved acid initially, a weak acid will yield significantly less hydronium ions than a strong acid.

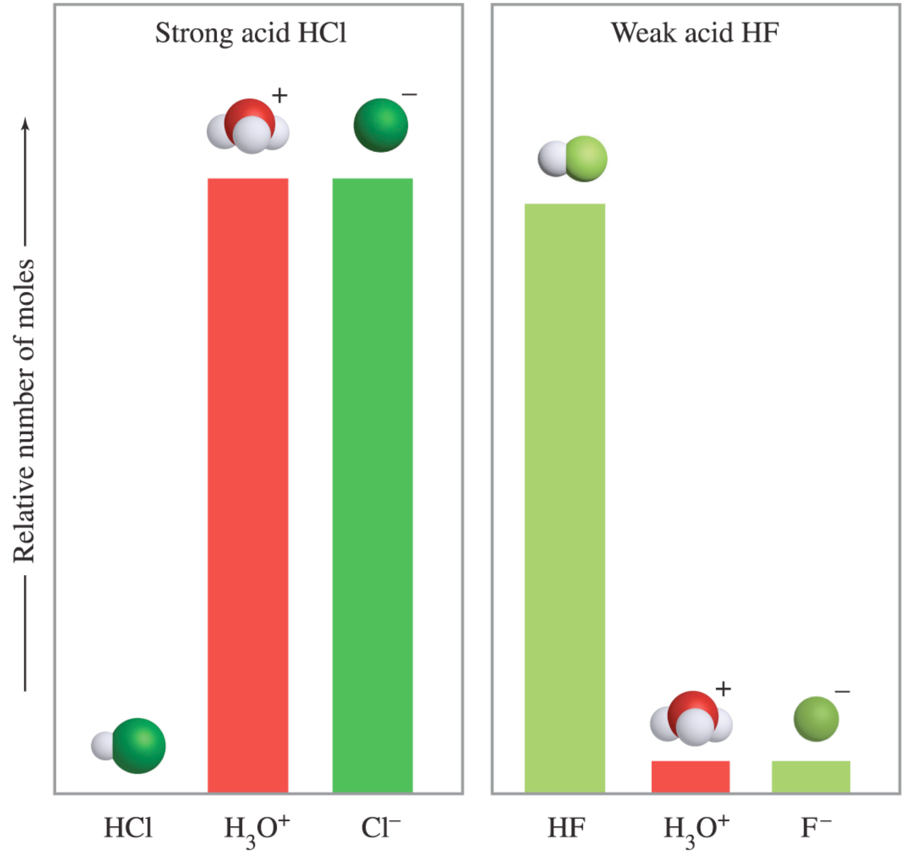


FIGURE 16.1

Ionization Reactions of Acids and Bases

16.1

A <u>weak base</u> is a weak electrolyte and dissociates/*ionizes incompletely* in water.

- As the ionization is *incomplete* (often <10 %) this is an ***equilibrium*** process.

only Na^+, OH^-
no NaOH!

strong base $\quad NaOH \xrightarrow{100\%} Na^+ + OH^-$ $\quad K_b \gg 1$

mostly NH_3!
few NH_4^+, OH^-

weak base $\quad NH_3 + H_2O \overset{<10\%}{\rightleftharpoons} NH_4^+ + OH^-$ $\quad K_b < 1$

- Given the same number of moles of dissolved base initially, a weak base will yield significantly less hydroxide ions than a strong base.

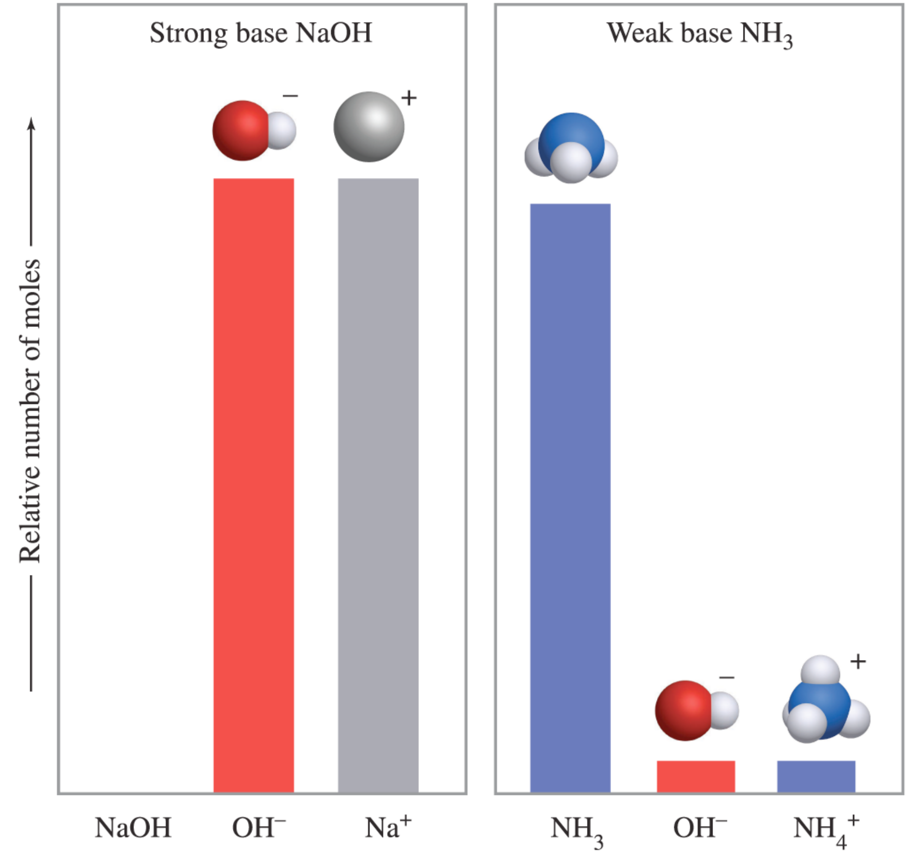


FIGURE 16.2

Weak Acids and Bases

16.5 (abbreviated)

As the ionization of weak acids and weak bases are *equilibria*, the magnitudes of their ionization equilibrium constants (K_a or K_b) is an indication of their relative **strengths**.

Acid ionization constant, K_a

$$HA(aq) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + A^-(aq)$$

$$K_a = \frac{[H_3O^+][A^-]}{[HA]}$$

"a" for acid

larger K_a, stronger acid

Base ionization constant, K_b

$$B(aq) + H_2O(\ell) \rightleftharpoons HB^+(aq) + OH^-(aq)$$

$$K_b = \frac{[HB^+][OH^-]}{[B]}$$

"b" for base

larger K_b, stronger base

IN-CLASS QUESTION 2

Learning Objective(s): 16.1, 16.4

For the following acids and bases, identify the correct arrow needed to complete each of the ionization reactions in water. Then indicate whether the corresponding values of K_a or K_b are expected to be large or small.

	In the box, write $\rightarrow$ or $\rightleftharpoons$ to complete the chemical equation.	Circle one answer per row.	
strong acid	$HClO_4(aq) + H_2O(\ell)$ $\boxed{\longrightarrow}$ $H_3O^+(aq) + ClO_4^-(aq)$	$K_a \gg 1$	$K_a < 1$
weak acid	$HF(aq) + H_2O(\ell)$ $\boxed{\rightleftharpoons}$ $H_3O^+(aq) + F^-(aq)$	$K_a \gg 1$	$K_a < 1$
strong base	$Ca(OH)_2(aq) + H_2O(\ell)$ $\boxed{\longrightarrow}$ $2OH^-(aq) + Ca^{2+}(aq)$	$K_b \gg 1$	$K_b < 1$
weak base	$NH_3(aq) + H_2O(\ell)$ $\boxed{\rightleftharpoons}$ $OH^-(aq) + NH_4^+(aq)$	$K_b \gg 1$	$K_b < 1$

Autoionization of Water

16.3

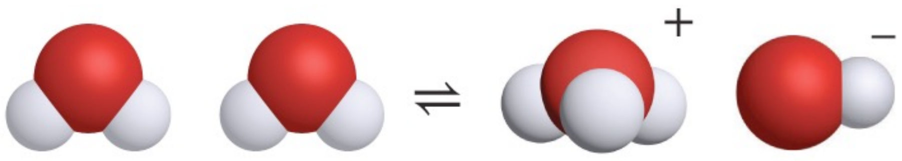
Water can act as *both* an acid and a base.

- Water molecules can undergo an acid-base reaction with themselves. This is called autoionization.
- The equilibrium constant for the autoionization process is denoted $\boldsymbol{K_w}$.
- Adding acids or bases will increase $[H_3O^+]$ or $[OH^-]$, but the system will always re-establish equilibrium, which is defined by the value of $\boldsymbol{K_w}$.

$$H_2O(l) + H_2O(l) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$$

$$K_w = [H_3O^+][OH^-] = 1.00 \times 10^{-14}$$

"w" for water



pH Calculations

16.4

The pH scale is a convenient means of conveying the hydronium ion concentration.

The pOH scale is a convenient means of conveying the hydroxide ion concentration.

$pH = -\log[H_3O^+]$ $pOH = -\log[OH^-]$ $pH + pOH = 14$ $K_w = [H_3O^+][OH^-] = 1.00 \times 10^{-14}$

- Solutions *often* have pH (or pOH) between 1 and 14, but not always.
- At 25 °C: pH < 7 Acidic solution $[H_3O^+] > [OH^-]$

  pH = 7 Neutral solution $[H_3O^+] = [OH^-]$

  pH > 7 Basic solution $[H_3O^+] < [OH^-]$

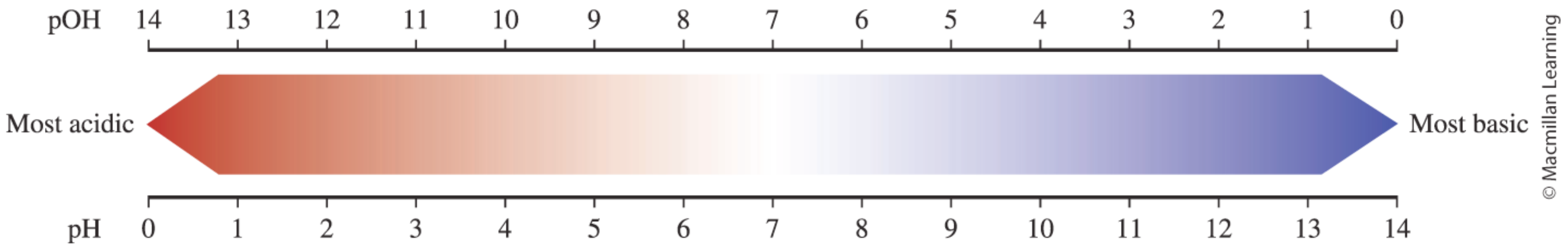


FIGURE 16.4

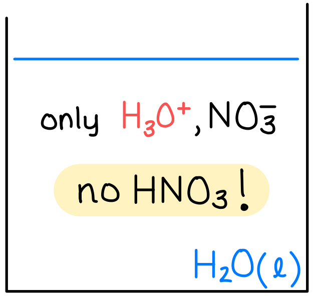
EXAMPLE PROBLEM

Learning Objective(s): 16.2, 16.5

Consider a 1.57×10^{-3} M $HNO_3(aq)$ solution at 25 °C.

Calculate the pH, pOH, $[H_3O^+]$, and $[OH^-]$ for this solution.

For a strong acid, 100% ionization, so $HNO_3(aq) + H_2O(\ell) \xrightarrow{100\%} H_3O^+(aq) + NO_3^-(aq)$



$$[H_3O^+] = \frac{1.57 \times 10^{-3} \text{ mol } HNO_3}{1 \text{ L}} \times \frac{1 \text{ mol } H_3O^+}{1 \text{ mol } HNO_3} = \frac{1.57 \times 10^{-3} \text{ mol } H_3O^+}{1 \text{ L}} = 1.57 \times 10^{-3} \text{ M } H_3O^+$$

$$pH = -\log[H_3O^+] = -\log(1.57 \times 10^{-3}) = 2.804$$

$$pOH = 14 - pH = 14 - 2.804 = 11.196$$

$$[OH^-] = 10^{-pOH} = 10^{-11.196} = 6.37 \times 10^{-12} \text{ M}$$

alternatively, $[OH^-] = \dfrac{K_w}{[H_3O^+]} = \dfrac{1.00 \times 10^{-14}}{1.57 \times 10^{-3}} = 6.37 \times 10^{-12}$ M

$$K_w = [H_3O^+][OH^-] = 1.00 \times 10^{-14}$$

$[H_3O^+] \rightarrow pH$: $pH = -\log[H_3O^+]$

$pH \rightarrow [H_3O^+]$: $[H_3O^+] = 10^{-pH}$

$pH + pOH = 14$

$pOH \rightarrow [OH^-]$: $[OH^-] = 10^{-pOH}$

$[OH^-] \rightarrow pOH$: $pOH = -\log[OH^-]$

EXAMPLE PROBLEM

Learning Objective(s): 16.4, 16.5

Order the solutions (of equal concentration) from lowest in pH to highest in pH.

Hint: Answer this question without explicitly calculating the pH values.

strong acid **A**	strong base **B**	weak base **C**	weak acid **D**	strong base **E**
$HBr(aq)$	$KOH(aq)$	$NH_3(aq)$	$HCN(aq)$	$Ca(OH)_2(aq)$

$pH = -\log[H_3O^+]$ $\quad$ $pH + pOH = 14$

$pOH = -\log[OH^-]$ $\quad$ $K_w = [H_3O^+][OH^-] = 1.00 \times 10^{-14}$

A and D are acids $\longrightarrow$ pH lower than B, C, E (bases) $\longrightarrow$ pH_A and pH_D < pH_C, pH_B, pH_E

$\longrightarrow$ A is a stronger acid than D $\longrightarrow$ $[H_3O^+]_A > [H_3O^+]_D$ $\longrightarrow$ $pH_A < pH_D$

B and E are strong bases $\longrightarrow$ E yields twice as much OH^- as B $\longrightarrow$ $[OH^-]_E > [OH^-]_B$ $\longrightarrow$ $pOH_E < pOH_B$ $\longrightarrow$ $pH_E > pH_B$

C is a weak base $\longrightarrow$ C yields less OH^- than B and E $\longrightarrow$ $[OH^-]_E > [OH^-]_B > [OH^-]_C$ $\longrightarrow$ $pOH_E < pOH_B < pOH_C$ $\longrightarrow$ $pH_E > pH_B > pH_C$

Highest $[H_3O^+]$	A	>	D	>	C	>	B	>	E	*Lowest $[H_3O^+]$*
Lowest in pH	A	<	D	<	C	<	B	<	E	*Highest in pH*

BONUS QUESTION

Assessed on: Summer 2021, Final Exam

Select all the **true** statements.

- ❑ All initial rates are examples of instantaneous rates.
- ❑ It is possible for molecules to collide with the correction orientation without resulting in a chemical reaction.
- ❑ For a chemical reaction at equilibrium, the amounts of reactants and products are equal.
- ❑ The values of Q and K for a given reaction can never be the same.
- ❑ The majority of strong acid molecules remain intact (i.e., in their protonated forms) in aqueous solution.